Quadratic-width and linear-depth learning bounds for Bayesian networks with stable populations

ChatGPT-6 Astra Claude Fable 5 and Opus 5 Kaarel Hänni
Dmitry Vaintrob*

Research draft, 29 October 2026

Abstract

We prove sample bounds comparable to the parameter count of a small teacher, independently of arbitrarily larger learner width, for exact Bayesian learning with Gaussian tanh priors restricted by stability and redundancy of neuron populations. For a Boolean teacher of width C and $L-1$ gate stages, a digital-population condition gives sufficient samples $O(dC+LC^2)$ at fixed calibration, accuracy, and confidence, with no hidden logarithm of C or L. The condition requires that averaging neurons preserve predictions, that the averaged hidden states be nearly Boolean with a fixed relative margin at every gate, and that the number of populations be calibrated by readout energy. It is specified without knowing the target or its complexity. Every eligible Boolean circuit has a positive-Gaussian-mass implementation satisfying it. A condition allowing analog hidden states instead requires internal-noise stability to persist after averaging; it gives $\widetilde{O}(dC+LC^2)$, with a stronger per-layer version giving $O(C^2L\log^5(2+C))$ at fixed other resources. All guarantees hold simultaneously at every learner width $N \geq C$ and every positive mean-field-or-stronger Gaussian scale, including finite mean field. These are different structural prior restrictions, not an improvement for unrestricted networks or for internal-noise stability alone. The proofs do not establish that ordinary trained networks satisfy the conditions, and are not Lean formalized.

1 Introduction

A dense network with width C and depth L has roughly $dC+LC^2$ parameters. A Bayesian learner may have arbitrarily larger width, so counting its own parameters does not explain learning a target with a small implementation. A useful statistical goal is a sample bound comparable, up to logarithms, to the parameter count of the smaller implementation.

We obtain such a bound for Boolean-circuit targets by restricting the prior in three ways. The network must admit averaging into a limited number of neuron populations; this averaging must preserve its clipped predictions; and the averaged network itself must tolerate finite simultaneous internal disturbances. The population budget is calibrated by the network's own readout energy and the architecture, not by the unknown target complexity. The restriction has positive Gaussian probability and admits every Boolean teacher considered below.

The new stability requirement matters for the depth dependence. A generic parameter cover propagates rounding errors through all layers and therefore needs an exponentially fine grid. Here the rounded network is viewed instead as the unrounded network with an explicitly constructed disturbance after each activation. Tanh bounds the rounded hidden states, so the total required

*Authors are listed in alphabetical order.

disturbance is polynomially small in the grid mesh and depth. Finite-radius robustness then controls the final clipped output without multiplying layer norms.

There are three precise versions of the result:

1. Robustness of the averaged representation on the usual aggregate noise ball gives the nearly parameter-count bound of Theorem 1.
2. Allowing each layer its full calibrated noise budget gives a bound literally linear in depth, with only width logarithms (Theorem 8).
3. Saturation and relative gate margins permit a constant-precision digital code. With a stronger population-energy calibration this gives literal quadratic width and linear depth (Theorem 9). The digital and analog conditions are different support restrictions; neither is asserted to follow from the other.

The first construction counts dense compressed-network parameters. The last counts constant-radius nets for normalized gate rows and avoids weight-magnitude and precision costs altogether. The Boolean target is a bounded-fan-in circuit; the quadratic benchmark is sufficient, not a claim that quadratic dependence is necessary.

The theorems include actual finite mean-field priors and all stronger scales. The same training-sample event works at every eligible learner width and scale. Neither an infinite-width approximation nor an efficient posterior-sampling algorithm is asserted. Earlier population-averaging arguments motivate the model and its Gaussian estimates [1]; all estimates needed for the present theorem are proved below.

2 Learner, teacher, and prior restriction

Let $d \geq 1$, $L \geq 2$, and $H = L - 1$. Inputs lie in $\mathcal{X} = \{-1, 1\}^d$, with an arbitrary probability distribution D. Write $\bar{x} = (x, 1)$. A network of hidden width N has

$$h^1 = \tanh(W^1 \bar{x}), \qquad h^s = \tanh(W^s h^{s-1}) \quad (2 \leq s \leq H), \qquad z_w(x) = v^\top h^H.$$

There are no additional hidden biases; a constant input coordinate and counted constant wires supply them in Boolean implementations. The raw Gaussian law $\mu_{N,t}$ has independent centered entries with variances

$$\frac{t^2}{d} \text{ in } W^1, \qquad \frac{t^2}{N} \text{ in } W^s \ (s \geq 2), \qquad \frac{t^2}{N^2} \text{ in } v, \qquad 0 < t \leq 1. \tag{1}$$

The scale may be prescribed as a function $t = t(d, L, N)$. In particular, $t(N) = N^{(1-\gamma)/2}$ gives every $\gamma \geq 1$ in the usual exponent family. Define the readout energy and clipped score by

$$E_v(w) = N \|v\|_2^2, \qquad T(u) = \mathrm{clip}(u, -1/2, 1/2).$$

Population averaging. A balanced partition into q populations has group sizes in $\{\lfloor N/q \rfloor, \lceil N/q \rceil\}$. For each hidden layer choose such a partition and let P_s be its orthogonal averaging projection. The averaged network $\bar{w}$ has

$$\bar{W}^1 = P_1 W^1, \qquad \bar{W}^s = P_s W^s P_{s-1} \ (s \geq 2), \qquad \bar{v} = P_H v. \tag{2}$$

Its clean hidden activations are constant within populations. Consequently it computes exactly the same function as a network of width q obtained by collecting equal neurons. Notice that $E_v(\bar{w}) \leq E_v(w)$.

Internal disturbances. For a network u of width N, add a vector η^s after each tanh activation and a scalar η^{out} at the output. Denote its output by $z_u(x;\eta)$. The disturbance norm and calibrated radius are

$$\|\eta\|_*^2 = \sum_{s=1}^{H} \frac{\|\eta^s\|_2^2}{N} + |\eta^{\text{out}}|^2, \qquad r_c(u) = \frac{c}{\sqrt{1+HE_v(u)}}, \qquad c > 0. \tag{3}$$

The disturbed states need not lie in $[-1,1]^N$.

The support condition. Fix $\lambda \geq 1$ and $c > 0$ before observing labels. Let $\mathcal{S}_{N,\lambda,c}$ consist of those networks for which some $q \leq N$ and balanced partitions satisfy all three conditions:

$$q \leq 1 + E_v(w)/\lambda, \tag{4}$$

$$\sup_{x\in\mathcal{X}} |T(z_w(x)) - T(z_{\bar{w}}(x))| \leq 1/16, \tag{5}$$

$$|T(z_{\bar{w}}(x;\eta)) - T(z_{\bar{w}}(x))| \leq 1/32 \quad (x \in \mathcal{X},\ \|\eta\|_* \leq r_c(\bar{w})). \tag{6}$$

In (6), $\bar{w}$ denotes the averaged network in (2). Thus the radius is calibrated by the averaged network's *own* readout energy. The partitions may depend on the weights, but not on the training labels.

The support is Borel. There are finitely many partition choices and inputs; for each choice the noise supremum is over a compact ball whose radius depends continuously on the weights. Equivalently one may rescale to a fixed unit ball and use a countable dense set.

The learner first restricts $\mu_{N,t}$ to this support and then, for a sample $S = ((X_i, f(X_i)))_{i=1}^n$, conditions on

$$f(X_i)z_w(X_i) > 1/2 \quad (1 \leq i \leq n). \tag{7}$$

Write the posterior as $Q_{S,N,t}$. Two further restrictions may optionally be retained: global clean margin $|z_w(x)| > 1/2$ at every input, and the original-network internal stability condition

$$|T(z_w(x;\eta)) - T(z_w(x))| \leq 1/16 \quad (x \in \mathcal{X},\ \|\eta\|_* \leq r_c(w)). \tag{8}$$

The theorem holds for either, both, or neither of these optional restrictions. When global margin is imposed, (7) is equivalent to conditioning on the exact training signs.

Teacher convention. The target is computed by a single-output, depth-H, width-C layered Boolean circuit with NAND, COPY, and NOT gates. A reserved constant-one wire at every stage, including the final stage, is included in C, and $C \geq 2$. Compilation from a different basis must count its width and depth overhead. A circuit of smaller logical depth can be padded with COPY gates. Thus L is the actual learner depth and the common depth after teacher padding; it cannot be replaced by a shallower teacher's intrinsic depth while leaving learner depth unspecified. This comparator is a Boolean circuit, not an arbitrary real-weight neural teacher.

3 An explicit learning theorem

Define target constants

$$
\begin{aligned}
e_* &= \frac{1}{16\sqrt{2C}}, & \tau_0 &= e_*/4, \\
b_L &= \sqrt{16\log 9 + 8\log(4L)}, & A &= \max\{32b_L, 2\log(8/e_*)\}, \\
a &= \max\{4\sqrt{\lambda}, 256c\}, & \rho &= \min\{1/4, A/[4(d+1)], Ae_*/(64\sqrt{C})\}, \\
q_0 &= d+2+(H-1)C, & B_0 &= 9dA^2 + 18(H-1)CA^2 + 2a^2C, \\
D_* &= B_0 + q_0 \log\frac{e\sqrt{2\pi}}{2\rho} + \log 2. & & \qquad (9)
\end{aligned}
$$

For $0 < \varepsilon, \delta < 1$, set

$$
\begin{aligned}
B &= \max\{8, 8[D_* + \log(4/\varepsilon)]\}, & Q &= \lceil 1 + B/\lambda \rceil, \\
J &= 8\left[D_* + \log(4/\varepsilon) + (H+1)\log Q + \frac{\log 2}{2}\{d+2+(H-1)Q\}\right], & & \\
R &= 1 + \sqrt{QJ}, & P_Q &= Q(d+2) + (H-1)Q^2, \\
\zeta &= \min\left\{\frac{1}{32}, \frac{c}{2\sqrt{(1+HB)((d+1)^2+HQ^2)}}\right\}. & & \qquad (10)
\end{aligned}
$$

Theorem 1 (Nearly parameter-count learning)**.** *Let f have the counted Boolean implementation above. If*

$$
n \geq \frac{2}{\varepsilon}\left[P_Q \log(3 + 4R/\zeta) + \log(1/\delta)\right], \qquad (11)
$$

then, with probability at least $1-\delta$ over iid $X_i \sim D$,

$$
\mathbb{E}_{w\sim Q_{S,N,t}} D\{x : f(x)z_w(x) \leq 1/4\} \leq \varepsilon \qquad (12)
$$

simultaneously for every integer $N \geq C$ and every $0 < t \leq 1$. All posterior normalizers are positive. The same conclusion holds with any choice of the optional restrictions described above.

Corollary 2 (Joint width–depth rate)**.** *Take $\lambda = d + L$. For fixed c, ε, δ, the sufficient sample bound in Theorem 1 is*

$$
\widetilde{O}(dC + LC^2),
$$

where the soft-O suppresses logarithmic factors in d, C, L. More precisely, at fixed $d, c, \varepsilon, \delta$, put

$$
F = 1 + \log^2(2+C) + \log(2+L), \qquad G = \log(2+CL).
$$

Then a sufficient bound is $O(C^2LF^2G)$. In particular its fixed-width depth dependence is $O(L\log^3(2+L))$, and its fixed-depth width dependence is $O(C^2\log^5(2+C))$.

Remark 3 (Prediction and limiting conventions)**.** *The posterior-average sign error is no greater than the margin error in (12). If $p_S(x) = Q_{S,N,t}\{f(x)z_w(x) > 0\}$, then $D\{p_S \leq 2/3\} \leq 3\varepsilon$ and $D\{p_S \leq 1/2\} \leq 2\varepsilon$. Replacing ε by $\varepsilon/3$ in the budget gives strict correct-label probability above $2/3$ outside test mass ε. The theorem covers all $\gamma \geq 1$ by the substitution following (1), and arbitrarily small positive scales at each fixed width. It also bounds the corresponding limsup risk as $t \downarrow 0$. If parameter posteriors have a weak limit, its sign-error bound follows by Portmanteau using the open event $fz_w < 1/4$. This observation does not claim that the closed margin event $fz_w \leq 1/4$ automatically transfers unchanged at boundary mass.*

4 A positive-mass robust Boolean implementation

Lemma 4 (Target cylinder)**.** *For every $N \geq C$ and $0 < t \leq 1$, there is a set of weights of raw Gaussian probability at least $\exp(-ND_*/t^2)$ which belongs to $\mathcal{S}_{N,\lambda,c}$, satisfies both optional restrictions, and has $f(x)z_w(x) > 1$ at every input.*

Proof. Partition the neurons into C balanced clone groups of sizes k_b, so $C/2 \leq N/k_b \leq 2C$. Fill unused logical slots with constants or copies. The ideal state $b^s(x) \in \{-1,1\}^N$ is constant on each group and has norm $\sqrt{N}$. Let P_s average these groups.

Give every gate gain A. In the $\{-1,1\}$ convention NAND uses the signed affine expression $1-u-v$, COPY uses u, and NOT uses $-u$. The counted constant wire supplies the first term inside hidden layers. Every ideal signed preactivation is at least A, and at most three coefficients of magnitude A enter a gate. Divide each coefficient between two logical wires equally over the source clones. Let the readout average the output-wire clones with total coefficient a. Call these weights w_*.

Use standardized coordinates $\sqrt{d}W^1$, $\sqrt{N}W^s$, and Nv, whose raw Gaussian variances are all t^2. Restrict each first-layer coordinate, each internal row's coordinate along each of the C normalized clone-group directions, and each readout coordinate to within ρt of its teacher value. Every internal matrix can then be written

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P_{s-1}, \qquad \|R^s\|_{\mathrm{op}} \leq \rho t\sqrt{C}, \qquad E^s P_{s-1} = 0.$$

The unconstrained Gaussian matrices E^s are independent of these coordinate restrictions. Two $1/4$-nets on the Euclidean unit sphere, each of size at most 9^N, give

$$\mathbb{P}\{\|E^s\|_{\mathrm{op}} > b_L t\} \leq 2\exp(2N\log 9 - Nb_L^2/8) \leq \frac{1}{2L}.$$

Indeed, a fixed bilinear form has variance at most t^2/N, and twice the largest value over the nets bounds the operator norm. Hence all the bulk norm bounds hold with conditional probability at least $1/2$.

We verify a stronger disturbance invariant than needed: allow RMS magnitude at most τ_0 separately at each hidden layer, and an output disturbance at most τ_0. Write $e_s = \left\|h^s_\eta - b^s\right\|_2/\sqrt{N}$. If $e_{s-1} \leq e_*$, every clone-group mean differs from its ideal bit by at most $e_*\sqrt{N/k_b} \leq 1/16$. The intended signed gate preactivation therefore remains at least $A/2$. For $\xi \in \{-1,1\}$ the elementary inequality

$$\xi q \geq A/2 \quad \Longrightarrow \quad |\tanh(q+u) - \xi| \leq 2e^{-A/2} + 8|u|/A$$

holds: if $|u| < A/4$, use the tanh tail bound, and otherwise the right side is at least two. Since $E^s b^{s-1} = 0$ and $\left\|h^{s-1}_\eta\right\|_2/\sqrt{N} \leq 1+\tau_0$, it follows that

$$e_s \leq 2e^{-A/2} + (8/A)\rho\sqrt{C}(1+\tau_0) + (8b_L/A)e_{s-1} + \tau_0 \leq e_*.$$

Each summand is at most $e_*/4$ by (9). The first-layer preactivation perturbation is at most $\rho(d+1) \leq A/4$, which initializes the same invariant. At the readout,

$$f(x)z_w(x;\eta) \geq 15a/16 - \rho(1+\tau_0) - \tau_0 > 1.$$

The readout center and perturbation satisfy $\sqrt{N}\,\|v_*\|_2 \geq a\sqrt{C/2}$ and $\sqrt{N}\,\|v - v_*\|_2 \leq \rho$. Thus

$$E_v(w) \geq a^2 C/8 \geq 2\lambda C, \qquad r_c(w) \leq \frac{2\sqrt{2}c}{a\sqrt{C}} \leq \tau_0. \tag{13}$$

This proves original-network stability (8) with zero clipped variation, as well as the global margin and count condition.

The clone projections map this cylinder into itself. Averaging rows preserves the constrained intervals because teacher rows agree within each group, and right projection kills E^s. First-layer and readout intervals are preserved as well. Consequently the averaged network has the same noisy margin invariant, and its own readout energy satisfies (13). Both its clean and noisy clipped scores equal $f(x)/2$ throughout its calibrated noise ball. This proves (6), even with zero in place of $1/32$, and proves (5) with zero error.

There are Nq_0 independent constrained coordinates; the sum of the squared centers is at most NB_0. For a scalar $Z \sim \mathcal{N}(0, t^2)$ and $\rho \le 1/4$,

$$\mathbb{P}\{|Z - m| \le \rho t\} \ge \frac{2\rho}{e\sqrt{2\pi}} e^{-m^2/t^2}.$$

This follows by bounding the density on the interval and using $(|m|/t+\rho)^2/2 \le m^2/t^2+\rho^2$. Multiply these estimates and the independent bulk-event probability $1/2$. Since $t \le 1$ and $N/t^2 \ge 1$, the product is at least $\exp(-ND_*/t^2)$. □

5 Posterior control of adaptive population choices

For a fixed partition tuple, let $k_{s,a}$ be its group sizes. The exact width-q quotient has coefficients

$$\begin{aligned} U^1_{aj} &= k_{1,a}^{-1} \sum_{i \in a} W^1_{ij}, \\ U^s_{ab} &= k_{s,a}^{-1} \sum_{i \in a,\ j \in b} W^s_{ij}, \\ u_b &= \sum_{j \in b} v_j. \end{aligned}$$

Define its energy and parameter count by

$$\kappa_q = \frac{d}{q} \left\| U^1 \right\|_F^2 + \sum_{s=2}^{H} \|U^s\|_F^2 + q \|u\|_2^2, \qquad P_q = q(d+2) + (H-1)q^2.$$

Lemma 5 (A partition-uniform Gaussian tail)**.** *For every integer $Q \ge 1$ and $J > 0$,*

$$\begin{aligned} &\mu_{N,t}\{\textit{some balanced partition tuple with } q \le \min(N, Q) \textit{ has } \kappa_q > J\} \\ &\le Q \exp\left(HN \log Q + \frac{N}{2}[d + 2 + (H-1)Q] \log 2 - \frac{NJ}{8t^2} \right). \end{aligned}$$

Proof. For fixed partitions, the quotient coefficients are independent centered Gaussians, with variances respectively

$$\frac{t^2}{dk_{1,a}}, \qquad \frac{t^2 k_{s-1,b}}{N k_{s,a}}, \qquad \frac{t^2 k_{H,b}}{N^2}.$$

After the weights in κ_q are applied, every squared Gaussian coordinate has variance at most $2t^2/N$. Balance gives $N/(qk) \le 2$, $qk/N \le 2$, and ratios of group sizes at most two. Therefore

$$\mathbb{E} \exp\left(\frac{N\kappa_q}{8t^2} \right) \le 2^{P_q/2}, \qquad \mathbb{P}(\kappa_q > J) \le e^{-NJ/(8t^2)} 2^{P_q/2}.$$

There are at most q^{HN} ordered partition tuples. If $q \le N$, then $P_q/N \le d + 2 + (H-1)q$. Sum the bound over $q \le \min(N, Q)$ and use the upper bound Q on that sum's length. This controls partitions before any weight-dependent selection. □

Lemma 6 (Posterior truncation)**.** *Outside posterior mass at most $\varepsilon/2$, every network has $E_v(w) \le B$ and a witnessing partition with $q \le Q$ and $\kappa_q \le J$.*

Proof. For every correctly labeled sample, Lemma 4 lower-bounds the raw probability of all prior restrictions and training constraints by e^{-ND_*/t^2}. Under the raw Gaussian, NE_v/t^2 is chi-square with N degrees of freedom. Its exponential moment gives, for $B \ge 8$,

$$\mu_{N,t}(E_v > B) \le e^{-NB/(4t^2)} 2^{N/2} \le e^{-NB/(8t^2)}.$$

After dividing by the conditioning probability, the definition of B bounds this posterior probability by $\varepsilon/4$. On its complement, (4) supplies $q \le Q$.

For the exceptional partition event of Lemma 5, use $t \le 1$ and $N \ge 1$ to bound every positive exponent term by N/t^2 times the corresponding term in $J/8$. Its raw probability is then at most

$$e^{-ND_*/t^2} e^{-(N/t^2)\log(4/\varepsilon)}.$$

Posterior normalization bounds it by $\varepsilon/4$. Off these two exceptional events every partition of size at most Q has energy at most J, so any witnessing partition can be chosen. □

6 Robust rounding without an exponential depth cost

Lemma 7 (Rounding as an internal disturbance)**.** *Every retained network from Lemma 6 has a candidate g in a fixed finite class $\mathcal{G}$, independent of N, t, such that*

$$\sup_{x \in \mathcal{X}} |T(z_w(x)) - T(g(x))| \le 3/32, \qquad \log|\mathcal{G}| \le P_Q \log(3 + 4R/\zeta).$$

Proof. Choose a witnessing partition and its quotient (U, u) of width $q \le Q$. The energy bound implies that every coefficient has magnitude at most $\sqrt{QJ} < R$. Round each coefficient to a grid in $[-R, R]$ containing zero, with error at most ζ and at most $3 + 4R/\zeta$ possible values. Let (U', u') denote the rounded quotient and h'^s its clean hidden trajectory. Tanh gives $|h'^s_a| \le 1$ at every depth.

The rounded trajectory is exactly the unrounded quotient's trajectory with the after-activation disturbances

$$\begin{aligned}
\eta^1_a &= \tanh((U'^1 \bar{x})_a) - \tanh((U^1 \bar{x})_a), \\
\eta^s_a &= \tanh((U'^s h'^{s-1})_a) - \tanh((U^s h'^{s-1})_a), \\
\eta^{\text{out}} &= (u' - u)^\top h'^H.
\end{aligned}$$

Lift each hidden disturbance constantly over its original population. This gives the same trajectory in the averaged width-N network, without any factor depending on population sizes. By tanh's Lipschitz bound,

$$|\eta^1_a| \le (d+1)\zeta, \qquad |\eta^s_a| \le q\zeta \ (s \ge 2), \qquad |\eta^{\text{out}}| \le q\zeta.$$

Hence the normalized total disturbance satisfies

$$\|\eta\|^2_* \le [(d+1)^2 + Hq^2]\zeta^2 \le \frac{c^2}{4(1+HB)} \le r_c(\bar{w})^2,$$

because $E_v(\bar{w}) \le E_v(w) \le B$. Applying the merged network's own robustness condition yields

$$\sup_x |T(z_{\bar{w}}(x)) - T(z_{U',u'}(x))| \le 1/32.$$

Together with (5), the total clipped approximation error is at most $3/32$.

Zero-pad the quotient to width Q. This preserves its function since $\tanh(0) = 0$, and it preserves the selected grid. There are P_Q physical parameters, so the class of all width-Q networks on this grid has the claimed log cardinality. Its grids and budgets depend only on the teacher resources, architecture, and statistical tolerances, not learner width or Gaussian scale. □

7 Generalization and the explicit rate

Proof of Theorem 1. Let ψ be one on $(-\infty, 11/32]$, zero on $[13/32, \infty)$, and affine between these values. Any retained, training-consistent w has a candidate $g \in \mathcal{G}$ with zero training losses $\psi(f(X_i)T(g(X_i)))$. Indeed, its correct clipped training score is $1/2$ and the approximation error is at most $3/32$. Whenever $f(x)z_w(x) \le 1/4$, the candidate's correct clipped score is at most $11/32$, so the ramp loss equals one.

For a fixed loss in $[0,1]$ of population mean r, the probability that all n iid losses vanish is at most $(1-r)^n \le e^{-nr}$: the probability of a positive loss is at least its mean. A union bound over $\mathcal{G}$ gives an event of probability at least $1-\delta$ on which every zero-training-loss candidate has population loss at most

$$\frac{\log|\mathcal{G}| + \log(1/\delta)}{n}.$$

This single event does not depend on N or t. On it, integrate the pointwise loss comparison over retained posterior networks and bound the exceptional posterior mass by $\varepsilon/2$. Lemmas 6 and 7 and (11) give

$$\mathbb{E}_Q D\{f z_w \le 1/4\} \le \varepsilon/2 + \frac{\log|\mathcal{G}| + \log(1/\delta)}{n} \le \varepsilon.$$

Lemma 4 proves positive normalizers for every stated choice of optional restrictions. □

Proof of Corollary 2. The target constants give

$$D_* = O((d + LC)\log^2(2dLC) + (\lambda + c^2)C).$$

Thus, with $\lambda = d + L$ and fixed c, ε, δ, $Q = \widetilde{O}(C)$ because $d + LC \le (d+L)C$. The quantities J, R, ζ^{-1} are polynomially bounded in d, C, L up to logarithms. Consequently the grid logarithm is $O(\log(2 + d + C + L))$ up to additional logarithms of these resources, and (11) gives $\widetilde{O}(dC + LC^2)$.

For the sharper fixed-d statement, the target gain has $A^2 = O(F)$ and $D_* = O(CLF)$. Hence

$$B = O(CLF), \quad Q = O(CF), \quad J = O(CLF), \quad R = O(C\sqrt{L}F), \quad P_Q = O(C^2LF^2).$$

The explicit mesh has a polynomial inverse in C, L, F; for example $\zeta^{-1} = O(1 + C^{3/2}L^{3/2}F^{3/2})$ at fixed d, c. Therefore $\log(3 + 4R/\zeta) = O(G)$. Multiplication gives $O(C^2LF^2G)$ and the stated specializations. □

8 Removing depth logarithms with per-layer stability

A stronger disturbance condition removes even the depth logarithms from Corollary 2, while continuing to allow analog hidden states. The condition is a product of noise balls: every layer may be perturbed by its full allowed amount simultaneously. We also use a Euclidean-ball parameter count, so that merely having more layers does not enlarge the required range of a typical coded coefficient.

Fix $\lambda \geq 1$ and $c > 0$. Retain the balanced-population count (4) and clean approximation (5), but replace (6) by

$$|T(z_{\bar{w}}(x;\eta)) - T(z_{\bar{w}}(x))| \leq \frac{1}{32} \quad \text{whenever} \quad \frac{\|\eta^s\|_2}{\sqrt{N}} \leq \tau_c(\bar{w}) \ (1 \leq s \leq H), \quad |\eta^{\text{out}}| \leq \tau_c(\bar{w}), \tag{14}$$

where

$$\tau_c(\bar{w}) = \frac{c}{\sqrt{1 + E_v(\bar{w})/\lambda}}.$$

This uses the averaged full-width network's own readout energy. It is a stronger disturbance requirement than the aggregate-radius condition: any vector of aggregate norm at most $r_c(\bar{w})$ obeys each bound above, since $H \geq 1$ and $\lambda \geq 1$. As before, the prior can optionally retain global margin and original-network stability (8). The training condition remains (7).

For this support, replace the target constants in (9) by

$$\begin{aligned} b &= \sqrt{16 \log 9 + 8 \log 4}, & A &= \max\{32b, 2\log(8/e_*)\}, \\ a &= \sqrt{\lambda}\max\{4, 256c\}, & D_{\text{pl}} &= B_0 + q_0 \log \frac{e\sqrt{2\pi}}{2\rho} + (H-1)\log 2. \end{aligned} \tag{15}$$

Here $e_*, \tau_0, \rho, q_0, B_0$ retain their previous formulas, evaluated with the new A, a. In particular the gate gain A no longer depends on depth. Define B, Q, J, P_Q by (10), with D_{pl} in place of D_*, and set

$$\tau_B = \frac{c}{\sqrt{1 + B/\lambda}}, \qquad \zeta_{\text{pl}} = \frac{\tau_B}{2\max\{d+1, Q\}}, \qquad K_{\text{ball}} = \sqrt{2\pi e}. \tag{16}$$

Theorem 8 (Linear depth under per-layer stability)**.** *Under the preceding support and teacher conventions, the conclusions of Theorem 1 hold if*

$$n \geq \frac{2}{\varepsilon}\left[P_Q \log\left(K_{\text{ball}}\left(1 + \frac{\sqrt{QJ/P_Q}}{\zeta_{\text{pl}}}\right)\right) + \log(1/\delta)\right]. \tag{17}$$

For $\lambda = d + L$ and fixed $d, c, \varepsilon, \delta$, this sufficient bound is

$$O(C^2 L \log^5(2 + C)).$$

In particular, there is no hidden logarithm of depth in this rate.

Proof. We first check the target cylinder. At each internal layer, the same Gaussian two-net bound as before now gives

$$\mathbb{P}\{\|E^s\|_{\text{op}} > bt\} \leq 2\exp(2N\log 9 - Nb^2/8) = 2 \cdot 4^{-N} \leq \frac{1}{2}.$$

The internal bulk matrices are independent of one another and of the constrained scalar coordinates. Thus all $H - 1$ bulk bounds hold with probability at least $2^{-(H-1)}$. Paying this probability directly,

rather than choosing a cutoff that makes their intersection probability at least $1/2$, removes the depth dependence from b and A. The scalar interval calculation in Lemma 4 gives raw cylinder probability at least $e^{-ND_{\mathrm{pl}}/t^2}$.

The same saturation induction proves signed output greater than one under RMS perturbation at most τ_0 separately at every hidden layer and output. Projection maps the cylinder into itself, so both w and $\bar{w}$ satisfy this invariant. For the averaged network,

$$E_v(\bar{w}) \geq a^2 C/8, \qquad \tau_c(\bar{w}) \leq \frac{c\sqrt{8\lambda}}{a\sqrt{C}} \leq \tau_0.$$

The new per-layer condition holds with zero clipped variation. Also $E_v(w) \geq 2\lambda C$, proving the population-count condition, and $a \geq 256c$ proves the optional original-network stability as before. The target normalizer bound is therefore valid for each stated prior.

Lemma 6 applies unchanged with this target cost. Outside posterior mass $\varepsilon/2$, there is a witnessing quotient of width $q \leq Q$, with $\kappa_q \leq J$, and $E_v(\bar{w}) \leq E_v(w) \leq B$. Hence its permissible per-layer radius is at least τ_B.

Round quotient coefficients to the nearest multiple of ζ_{pl}. The construction in Lemma 7 expresses the rounded trajectory exactly through population-constant internal disturbances. Every first-layer coordinate disturbance is at most $(d+1)\zeta_{\mathrm{pl}}$, and every subsequent hidden or output disturbance is at most $q\zeta_{\mathrm{pl}}$. Consequently each layer's RMS norm and the output disturbance are at most $\tau_B/2$. The per-layer condition bounds the rounded-to-averaged clipped error by $1/32$, and the total approximation error is again at most $3/32$.

It remains to count rounded coefficients without a depth-dependent range penalty. Since $d \geq 1$ and $q \leq Q$, the sum of squares of all unweighted quotient parameters is at most QJ. After zero-padding to width Q, their vector lies in a Euclidean ball in dimension P_Q of radius $S = \sqrt{QJ}$. Consider all lattice points produced by nearest-grid rounding of any vector in this ball. Their cubes of side ζ_{pl} have disjoint interiors and lie in the ball of radius $S + \zeta_{\mathrm{pl}}\sqrt{P_Q}$. For every positive integer P and radius R,

$$\mathrm{vol}(B_R) \leq \left(\sqrt{\frac{2\pi e}{P}}R\right)^P.$$

For example, integrate $\mathbf{1}_{\|x\|\leq R} \leq \exp(\alpha R^2 - \alpha\|x\|^2)$ and choose $\alpha = P/(2R^2)$. Dividing by the lattice-cube volume yields

$$|\mathcal{G}| \leq \left[K_{\mathrm{ball}}\left(1 + \frac{\sqrt{QJ/P_Q}}{\zeta_{\mathrm{pl}}}\right)\right]^{P_Q}.$$

Padding includes every $q \leq Q$ in the same ball, so no separate count of partitions or widths is needed at this step. The grid candidates need not themselves satisfy the prior restrictions.

The ramp and finite-class argument from Theorem 1 now give (17), on a single event for every $N \geq C$ and $0 < t \leq 1$.

For the rate, put $F_C = 1 + \log^2(2 + C)$ and fix $d, c, \varepsilon, \delta$. At $\lambda = d + L$, the modified target cylinder has

$$D_{\mathrm{pl}} = O(LCF_C), \qquad B = O(LCF_C), \qquad Q = O(CF_C), \qquad J = O(LCF_C).$$

More specifically, $D_{\mathrm{pl}} + \log(4/\varepsilon) \leq B/8$ and $B/\lambda \leq Q$. The formula for J therefore gives $J = O(LQ)$ at fixed dimension. If $H \geq 2$, then $P_Q \geq (H-1)Q^2$, so $QJ/P_Q = O(1)$ uniformly in depth. Also

$$\zeta_{\mathrm{pl}}^{-1} = \frac{2\max\{d+1, Q\}\sqrt{1 + B/\lambda}}{c} = O(Q^{3/2})$$

at fixed d, c. Thus the logarithm in (17) is $O(\log(2+C))$, while $P_Q = O(LC^2F_C^2)$. If $H = 1$, then $L = 2$ is fixed and the same claimed bound follows directly from the displayed budgets. This proves $O(C^2L\log^5(2+C))$. □

The two changes have distinct roles. Requiring simultaneous per-layer robustness makes the rounding precision independent of depth, and counting an energy ball rather than a coordinate cube removes a depth factor from the apparent coefficient range. Neither follows merely from clean population averaging. This theorem still permits analog population responses; the digital-margin condition introduced separately gives a stronger, literal C^2L bound.

9 A literal quadratic-width, linear-depth theorem

The logarithms in the preceding parameter-grid bound can also be removed under a different structural restriction: after merging, the network behaves as a collection of robust digital gates. This assumption is natural for a saturated implementation of a Boolean circuit, but is not claimed to hold for typical gradient-trained analog networks. The resulting prior still has positive probability around an implementation of every eligible Boolean circuit. Its support definition does not use that circuit or its width.

We retain the Gaussian tanh model, $H = L - 1$, the readout energy $E_v(w) = N\|v\|_2^2$, and the clipped score $T(u) = \max\{-1/2, \min\{u, 1/2\}\}$. For balanced population partitions into q groups at every hidden layer, let $\bar{w}$ be the population-averaged full-width network. Write its exactly equivalent width-q quotient as

$$h^0(x) = (x, 1), \qquad a^s(x) = U^s h^{s-1}(x), \qquad h^s(x) = \tanh(a^s(x)), \qquad z_{\bar{w}}(x) = u^\top h^H(x).$$

Thus U^1 has $d+1$ columns, subsequent U^s have q columns, and u has q coordinates.

Fix, before observing labels,

$$\lambda = (d+L)(1+\log(d+2)), \qquad \Psi(q) = q\log^2(2+q), \qquad \eta = \frac{1}{16}, \quad \kappa = \frac{1}{5}. \tag{18}$$

The event $\mathcal{D}_N$ consists of networks admitting such partitions, with $1 \le q \le N$, for which

$$\lambda\Psi(q) \le E_v(w), \tag{19}$$

$$\sup_x |T(z_w(x)) - T(z_{\bar{w}}(x))| \le \frac{1}{16}, \tag{20}$$

$$|h_a^s(x)| \ge \frac{15}{16}, \tag{21}$$

$$|a_a^s(x)| \ge \kappa\|U_{a,\cdot}^s\|_1, \qquad |z_{\bar{w}}(x)| \ge \kappa\|u\|_1. \tag{22}$$

The last two lines hold for every Boolean input, every hidden layer, and every indicated coordinate. All rows and the readout in (22) are required to be nonzero. The prior is the raw Gaussian conditioned on $\mathcal{D}_N$ and

$$\mathcal{G}_N = \{w : |z_w(x)| > 1/2 \text{ for every Boolean } x\}.$$

Training conditions further on the observed noiseless signs. We also allow intersection with the original internal-noise event $\mathcal{R}_{N,c}$, for any fixed $c > 0$: clipped scores must change by at most $1/16$ under every simultaneous post-activation and output disturbance whose norm

$$\left(\sum_{s=1}^{H} \|\xi^s\|_2^2/N + |\xi^{\text{out}}|^2\right)^{1/2}$$

is at most $c/\sqrt{1+HE_v(w)}$. This optional intersection does not change the conclusion below.

These are Borel restrictions involving finitely many inputs and partition tuples at each width. Saturation makes the populations nearly Boolean; the relative row margin asks that the sign of each gate tolerate a fixed fractional error in its aggregate incoming coefficients. The stronger population-count calibration in (19) is an additional choice of prior: it compensates for the logarithmic gains used in the Boolean implementation. It is not implied by the original robustness restriction.

9.1 Exact budgets and the learning statement

Let the Boolean target have layered width C, with H stages and the constant wires required by the NAND/COPY/NOT implementation counted. Put

$$\begin{aligned}
e_* &= \frac{1}{16\sqrt{2C}}, \qquad \tau_0 = e_*/4, \qquad b = \sqrt{16\log 9 + 8\log 4},\\
A &= \max\{32b, 2\log(8/e_*)\}, \qquad a = \sqrt{\lambda}\max\{4\log(2+C), 256c\},\\
\rho &= \min\{1/4, A/[4(d+1)], Ae_*/(64\sqrt{C})\},\\
q_0 &= d+2+(H-1)C,\\
B_0 &= 9dA^2 + 18(H-1)CA^2 + 2a^2C,\\
D_{\mathrm{dig}} &= B_0 + q_0\log\frac{e\sqrt{2\pi}}{2\rho} + (H-1)\log 2.
\end{aligned} \tag{23}$$

For $0<\varepsilon,\delta<1$, define

$$\begin{aligned}
B &= \max\{8, 8[D_{\mathrm{dig}} + \log(6/\varepsilon)]\},\\
Q &= \max\{q\in\mathbb{N} : q\ge 1,\ \Psi(q)\le B/\lambda\},\\
P_Q &= Q(d+2) + (H-1)Q^2.
\end{aligned} \tag{24}$$

The maximum exists and satisfies $Q\ge C$, as the proof verifies.

Theorem 9 (Literal quadratic width and linear depth)**.** *Under either prior just specified, all posterior normalizers are positive. If*

$$n \ge \frac{6}{\varepsilon}\left[\log Q + P_Q\log 33 + \log(1/\delta)\right], \tag{25}$$

then, with probability at least $1-\delta$ over iid training inputs, the posterior gives the correct label probability strictly greater than $2/3$ outside a set of test inputs of probability at most ε. The same sample event works for every $N\ge C$ and every $0<t\le 1$, under an arbitrary common training and test distribution on the Boolean cube.

There is an absolute constant K such that the simpler sufficient bound

$$n \ge \frac{K(1+c^2)^2}{\varepsilon}\left[(d+2)R + LR^2 + \log(1/\delta)\right], \qquad R = C + \frac{\log(6/\varepsilon)}{\lambda}, \tag{26}$$

implies the same conclusion. In particular, at fixed input dimension, calibration, accuracy, and confidence, the sufficient sample count is $O(C^2L)$, with no hidden logarithm of C or L.

The theorem covers the finite mean-field prior and every stronger scaling $\gamma\ge 1$, by taking $t(N) = N^{(1-\gamma)/2}$. Its digital hypothesis is separate from ordinary internal-noise stability; the conclusion also holds when both conditions are imposed. It strengthens clean population averaging through the stated digital and count requirements. It gives no claim about efficient posterior sampling or typical training outcomes.

9.2 A Gaussian cylinder inside the digital support

Lemma 10 (Digital target cylinder)**.** *For every $N \geq C$ and $0 < t \leq 1$, a set of raw probability at least $\exp(-ND_{\mathrm{dig}}/t^2)$ belongs to $\mathcal{G}_N \cap \mathcal{D}_N \cap \mathcal{R}_{N,c}$ and satisfies $f(x)z_w(x) > 1$ at every input.*

Proof. Use the balanced cloned Boolean implementation with gate gain A and readout coefficient sum a. In coordinates $\sqrt{d}W^1$, $\sqrt{N}W^s$, and Nv, constrain first-layer coordinates, every internal row's normalized clone-group directions, and all readout coordinates to intervals of radius ρt around their teacher values. The unconstrained internal blocks satisfy $E^s P = 0$. The same two-net Gaussian estimate used for the target cylinder gives

$$\Pr\{\|E^s\|_{\mathrm{op}} > bt\} \leq 2\exp(2N\log 9 - Nb^2/8) = 2 \cdot 4^{-N} \leq \frac{1}{2}.$$

The blocks are independent. Requiring all $H-1$ bulk bounds therefore costs at most a factor $2^{-(H-1)}$. This replaces a depth-dependent operator-norm cutoff by a constant cutoff; the probability cost is included in D_{dig}.

The saturation induction for the cloned construction gives hidden RMS error at most e_* relative to its ideal Boolean state, even with RMS noise at most τ_0 separately at every layer and output noise at most τ_0. Indeed its recurrence is

$$e_s \leq 2e^{-A/2} + (8/A)\rho\sqrt{C}(1+\tau_0) + (8b/A)e_{s-1} + \tau_0 \leq e_*.$$

The first-layer perturbation initializes the same invariant, and the signed readout is at least $15a/16 - \rho(1+\tau_0) - \tau_0 > 1$. There are Nq_0 independent constrained scalar coordinates with sum of squared centers at most NB_0. The scalar Gaussian interval estimate

$$\Pr\{|Z-m| \leq \rho t\} \geq \frac{2\rho}{e\sqrt{2\pi}} e^{-m^2/t^2} \quad (Z \sim \mathcal{N}(0,t^2))$$

and $N/t^2 \geq 1$ give the claimed probability $e^{-ND_{\mathrm{dig}}/t^2}$.

Averaging within clone groups maps this cylinder into itself. Both the original and averaged networks therefore have clipped score $f/2$, proving (20) and the global margin. Furthermore,

$$E_v(w) \geq a^2C/8 \geq 2\lambda C\log^2(2+C),$$

which proves (19). The calibration $a \geq 256c\sqrt{\lambda}$ also ensures that the original internal-noise ball lies in the preceding τ_0 product of balls, establishing the optional restriction.

It remains to check the digital conditions for the quotient. Its hidden coordinates are constant on groups of mass at least $1/(2C)$. The RMS error bound thus implies that every coordinate differs from its ideal bit by at most $\sqrt{2C}e_* = 1/16$, proving saturation. Every ideal gate row has ℓ_1 norm at most $3A$ and signed ideal preactivation at least A.

In the first layer, the quotient row perturbation has ℓ_1 norm at most $(d+1)\rho t/\sqrt{d} \leq A/4$. Signed preactivation divided by row norm is therefore at least $3/13 > 1/5$. For an internal row, the perturbation coefficient from group b is at most $\rho t\sqrt{k_b/N}$. Its total ℓ_1 norm is at most

$$\rho\sqrt{2C} \leq A/1024.$$

Accounting also for the coordinatewise $1/16$ error of the preceding state gives signed preactivation at least $831A/1024$ and row norm at most $3073A/1024$. Their ratio exceeds $1/5$. Finally, the quotient readout perturbation has ℓ_1 norm at most ρ, so its signed output is at least $15a/16 - \rho$ and its row norm at most $a + \rho$. As $a \geq 4$ and $\rho \leq 1/4$, their ratio is at least $14/17 > 1/5$. All these rows are nonzero. This verifies (22) and completes the support check. □

9.3 A constant-precision digital code

Lemma 11 (Finite digital class)**.** *For each integer $Q \geq 1$, the sign functions of networks in $\mathcal{G}_N \cap \mathcal{D}_N$ admitting witnesses with $q \leq Q$, over all learner widths N, belong to a fixed finite class $\mathcal{F}_Q$ with*

$$\log |\mathcal{F}_Q| \leq \log Q + P_Q \log 33.$$

This class does not depend on the sample, learner width, or Gaussian scale.

Proof. The unit ℓ_1 ball in $\mathbb{R}^m$ has an ℓ_1 net of radius $1/16$ with at most 33^m points. To see this, take a maximal $1/16$-separated set. Disjoint balls of radius $1/32$ around its points fit inside the ball of radius $1 + 1/32$; volume scaling gives the bound, and maximality gives a net.

For a quotient satisfying the digital conditions, let $b^s(x) = \mathrm{sign}(h^s(x))$. Normalize each nonzero row by its ℓ_1 norm. Since $\|h^s - b^s\|_\infty \leq 1/16$, replacing a preceding hidden state by its bits changes a normalized preactivation by at most $1/16$. Each hidden gate and the final readout on those bits consequently have the original sign with margin at least

$$\kappa - \eta = \frac{11}{80}.$$

The first-layer input is already Boolean, including its constant coordinate, so it has margin at least κ.

Replace every normalized row by a point in the fixed $1/16$ net for its dimension, and use threshold gates instead of tanh. On Boolean inputs this changes each preactivation by at most $1/16 < 11/80$. Induction through the layers preserves every hidden sign and the readout sign. At width q, the product of the row-net sizes is at most 33^{P_q}. Summing over $q \leq Q$ gives at most $Q33^{P_Q}$ possibilities. No count of adaptive population partitions is required: every resulting quotient is already covered by these same row nets.

Finally, the original global margin and (20) force the original and quotient scores to have the same sign at every input. Thus this code describes the actual posterior sign functions, not just an approximate surrogate. □

The finite nets may use real coefficients. The lemma counts a fixed finite function class; it does not claim a rational bit encoder or an efficient construction algorithm.

9.4 Posterior learning and the explicit polynomial

Proof of Theorem 9. Lemma 10 lower-bounds the raw conditioning probability by $e^{-ND_{\mathrm{dig}}/t^2}$ for every correctly labeled sample. Under the raw prior, NE_v/t^2 is chi-square with N degrees of freedom. For $B \geq 8$, its exponential moment gives

$$\mu_{N,t}\{E_v > B\} \leq e^{-NB/(8t^2)}.$$

After conditioning on prior support and training labels, the mass of this event is therefore at most

$$\exp\left[-\frac{N}{t^2}(B/8 - D_{\mathrm{dig}})\right] \leq \varepsilon/6.$$

Every remaining predictor has a witness with $q \leq Q$. The definition of Q is legitimate because $D_{\mathrm{dig}} \geq 2a^2C \geq 32\lambda\Psi(C)$, so $B/\lambda \geq 256\Psi(C)$.

Every retained sign function lies in $\mathcal{F}_Q$ and is consistent with the sample. A fixed function of population error greater than $\varepsilon/6$ is consistent with n iid labels with probability at most $e^{-n\varepsilon/6}$.

Lemma 11, a union bound, and (25) imply that, with probability at least $1-\delta$, every consistent member of $\mathcal{F}_Q$ has error at most $\varepsilon/6$. This event is independent of N, t. On it, the full posterior Gibbs error is at most $\varepsilon/3$: $\varepsilon/6$ for retained mass and at most $\varepsilon/6$ for discarded mass. The test mass where correct-label posterior probability is at most $2/3$ is at most three times this Gibbs error. This proves the stated confidence guarantee, including arbitrary posterior mixtures.

For the simpler envelope, all constants below are absolute. From (23),

$$A^2 \le K_1 \log^2(2+C), \qquad \rho^{-1} \le K_2(d+C+1), \qquad a^2 \le K_3(1+c^2)\lambda \log^2(2+C).$$

Using $d + LC \le (d+L)C$ and the dimension logarithm in (18), these inequalities give

$$D_{\mathrm{dig}} \le K_4(1+c^2)\lambda\Psi(C). \tag{27}$$

The interval logarithm contributes only $O(\log^2(2+C) + \log(d+2))$, and the bulk cost is linear in H; there is no logarithm of L in this estimate.

For $k \ge 1$, $\Psi(kC) \ge k\Psi(C)$. Moreover, for $q \ge kC$,

$$\Psi(q) \ge k\Psi(C) + (q - kC)\log^2 3.$$

Combining this with (27) and the definition of B bounds

$$Q \le K_5\left((1+c^2)C + \frac{\log(6/\varepsilon)}{\lambda}\right) \le K_5(1+c^2)R.$$

Since $\log Q \le Q$ and $P_Q \le (d+2)Q + LQ^2$, substitution in (25) proves (26) after enlarging one absolute constant. □

Corollary 12 (A polynomial accuracy envelope)**.** *For an absolute constant K, the sufficient sample condition*

$$n \ge \frac{K}{\varepsilon}\left[(1+c^2)^2(dC + LC^2) + \log^2(6/\varepsilon) + \log(1/\delta)\right] \tag{28}$$

implies every conclusion of Theorem 9.

Proof. Write $s = 1 + c^2$ and $v = \log(6/\varepsilon) > 1$. The preceding proof gives $Q \le K_5(sC + v/\lambda)$. Since $\lambda \ge d + L$,

$$(d+2)\frac{v}{\lambda} = O(v), \qquad L\frac{v^2}{\lambda^2} \le v^2, \qquad 2LsC\frac{v}{\lambda} \le 2sCv \le s^2C^2 + v^2.$$

Also $(d+2)sC = O(s^2dC)$, because $d \ge 1$ and $s \ge 1$. Consequently $\log Q + P_Q \log 33 = O(s^2(dC + LC^2) + v^2)$. Insert this estimate into (25). □

What the sharpening does and does not establish. The C^2L conclusion uses both digital relative margins and the stronger readout-energy population calibration. It does not assert the same bound for the original internal-noise prior, for clean neuron averaging alone, or for the unrestricted Gaussian prior. Every eligible Boolean circuit has a positive-mass witness, but that is an existence statement about an implementation, not evidence that generic trained networks satisfy the support condition. The proof is a conventional argument; no Lean formalization is claimed for this theorem.

10 Interpretation and limitations

The analog bounds are close, up to width logarithms, to the dense parameter count of a width-C, depth-L representation. The digital theorem removes those logarithms under a different structural condition and calibration. These are upper bounds from specified priors and proofs, not optimality theorems. A bounded-fan-in Boolean circuit can have a description of only $O(LC\log(C+d))$ bits, so quadratic width is not a universal lower bound for the target class.

The analog stability hypothesis has an operational meaning: a compressed or merged network must remain stable after it replaces the original model. Matching clean outputs alone does not ensure reliable behavior under numerical or internal perturbations. The Boolean witnesses have this property because saturated gates repeatedly restore approximate bits to their intended logical values. The support condition nevertheless remains strong. No claim is made that SGD or Adam typically produces it, or that an efficient algorithm finds its witnessing populations.

The original network's internal robustness alone does not supply the new argument. Clean-output closeness of two networks does not compare their noisy hidden trajectories. Requiring robustness of the averaged representation closes exactly this gap. The architecture-dependent count calibration $\lambda = d + L$ is also part of the changed prior; it is not a free improvement for the baseline population count. The digital theorem instead uses $\lambda = (d+L)(1+\log(d+2))$ and a logarithmically weighted population budget; its global output margin is needed to transfer the finite sign code back to the actual posterior predictor.

The proof uses bounded hidden activations at the rounding step. Extending this exact argument to an unbounded activation requires a suitable bound on the rounded trajectory or a separate control of large neuron activity. The theorem as stated is for tanh, and the target-mass proof uses its bit-restoring saturation. Finally, the result guarantees statistical performance of ideal Bayesian conditioning and does not provide an efficient inference algorithm.

Verification status. This manuscript gives conventional proofs and has not been formalized in Lean. Its main ingredients are adapted and sharpened from the population averaging analysis [1]; prior formal verification of related learning theorems does not certify this new support condition or its sample bound.

References

[1] ChatGPT-6 Astra, Claude Fable 5 and Opus 5, Kaarel Hänni, and Dmitry Vaintrob. *Quadratic-width learning bounds for Bayesian networks with stable neuron populations*. Research draft, 21 October 2026. Repository manuscript and source.